

SRI VASAVI ENGINEERING COLLEGE (Autonomous)
B. Tech VII Semester Regular Examinations, Nov/Dec-2026
(Model Paper-I)
DEEP LEARNING
(Computer Science & Engineering)

Time: 3 Hrs

Max. Marks: 70

		PART-A Answer All the Questions.	
1.			20 M
	i.	Define Artificial Intelligence.	CO1-K1(2M)
	ii.	List any four performance metrics used in Machine Learning.	CO1-K1(2M)
	iii.	What is a Perceptron?	CO2-K1(2M)
	iv.	State the purpose of an activation function	CO2-K1(2M)
	v.	Expand Keras and TensorFlow.	CO3-K1(2M)
	vi.	List the stages in a neural network model-building pipeline.	CO3-K1(2M)
	vii.	Define Convolutional Neural Network (CNN).	CO4-K1(2M)
	viii.	What is the Vanishing Gradient Problem?	CO4-K1(2M)
	ix.	What is Natural Language Processing (NLP)?	CO5-K1(2M)
	x.	Mention any two applications of Deep Learning.	CO5-K1(2M)
		PART-B All Questions Carry Equal Marks	
2.			10 M
	A.	i. Explain the differences between Machine Learning and Deep Learning with suitable examples.	CO1- K2(5M)
		ii. Discuss data preprocessing and train-validation-test split.	CO1- K2(5M)
		OR	
	B.	i. Explain the evaluation metrics Accuracy, Precision, Recall and F1-score.	CO1- K2(5M)
		ii. Discuss overfitting and underfitting with suitable examples.	CO1- K2(5M)
3.			10 M
	A.	i. Explain the architecture of Artificial Neural Networks.	CO2- K2(5M)
		ii. Explain the role of activation functions with neat diagrams.	CO2- K2(5M)
		OR	
	B.	i. Explain the complete training process of Deep Neural Networks including back propagation.	CO2- K2(5M)
		ii. Explain weight initialization Techniques.	CO2- K2(5M)
4.			10 M
	A.	i. Explain every stage from preprocessing to evaluation.	CO3- K3(5M)
		ii. Develop an ANN model using Keras for binary classification.	CO3- K3(5M)
		OR	
	B.	i. Build a multiclass classification model using Keras.	CO3- K3(5M)
		ii. Explain the multiclass model-building pipeline.	CO3- K3(5M)
5.			10 M
		i. Develop a CNN model using PyTorch for image classification and explain each layer involved.	CO4- K3(10M)
		OR	
	B.	i. Develop an RNN model using PyTorch for sequence prediction and explain the working principle.	CO4- K3(10M)
6.			10 M
	A.	i. Explain Deep Learning applications in Computer Vision	CO5- K2(5M)
		ii. Explain Deep Learning applications related to Natural Language Processing.	CO5- K2(5M)
		OR	
	B.	i. Explain Auto encoders, GANs, LSTMs with suitable examples.	CO5- K2(10M)
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